Antoni Kingston

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AntoniKingston

Education

MSc **Warsaw University of Technology (MiNI Faculty)**, Computer Science — Warsaw, Poland, Feb 2026 – present

BS **Warsaw University of Technology (MiNI Faculty)**, Data Science — Warsaw, Poland, Oct 2022 – Feb 2026
- *Adapting the library for classification utilizing discriminant analysis – gipsDA* (Bachelor's Thesis)

Experience

Center for Credible Artificial Intelligence, Research Intern — Warsaw, Poland, July 2026 – Aug 2026, 2 months
- Worked on research subject 'From Actions to Meaning: Do AI Models Build World Models'
- Created LLM world model building capabilities benchmarking paradigm based on cellular automata comprehension

Accenture, Data Warehouse Administrator (Internship) — Warsaw, Poland, July 2025 – Sept 2025, 3 months
- Analyzed long SQL queries to deal with reported inconsistencies in telecommunications DWH
- Developed a django web app for custom ticket management based on the system's API
- Got to know the inner workings of DWH operations, especially of ELT process

Projects

gipsDA R library
An R package for discriminant analysis classification using covariance matrices with permutation symmetries
- A fully functional R library available on CRAN, developed with best programming practices
- Enhances gips by estimating permutation symmetry shared by multiple matrices
- Provides Linear and Quadratic Discriminant Analysis classifiers utilizing gips methodology

Tweet Sentiment Predictor
Binary sentiment classification on 100k Sentiment140 tweets via NLP feature engineering and ensemble ML
- Built a three-stage Jupyter pipeline for feature engineering, selection, and modeling, with reproducible 5-fold cross-validation
- Engineered lexical, readability, temporal, and embedding features using VADER, TextBlob, Word2Vec, Sentence-BERT, and spaCy
- Selected predictors with L1 logistic regression and feature_engine; tuned XGBoost, LightGBM, ensembles, and a Keras MLP with Optuna, achieving ~81% cross-validated accuracy

A Simple GPT model
Generative language model training
- Implemented a causal Transformer language model in PyTorch with attention blocks and RMSNorm
- Built a custom BPE tokenizer and corpus-to-encoding pipeline; trained on Adam Mickiewicz works and generated text
- In progress: scaling training to the Polish Parliamentary Corpus (PPC)

Skills

Programming Languages: Python, Java

Mathematics: Linear ALgebra, Abstract Algebra, Differential Geometry, Probability Theory, Statistics

ML Frameworks: Pandas, Sklearn, PyTorch

Areas of Scientific Interests: Geometry of Concepts in the Latent Space of AI Models

Tools: bash, git, docker, latex, html, markdown

Languages: Polish (native), English (C2), German (B1)

Selected Achievements

- Won an episode of '1z10' TV quiz show
- Official WCA 5BLD solve (3rd place at 2021 Szansa Open)
- Cycled 408km in one go in a 24h cycling event